

Haptic Music Feedback through Audio Decomposition

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Abstract

This study presents a novel approach to improving musical haptic wearables by applying deep learning models for music source separation and pitch estimation. The system consists of a haptic vest and a pair of haptic gloves, designed to spatially convey musical elements throughout the body. By isolating instruments from an audio file and recognizing their pitches, our system provides intuitive haptic feedback that discretely represents each instrument's performance. Specifically, we map the piano to the gloves, the bass to the back of the vest, and the drums to the front of the vest. Then, we conducted a comparative study with a conventional audio-to-haptic method. The results showed that users experienced improved clarity, intuitiveness, and comprehension, indicating enhanced understanding of musical structure.

CCS Concepts

• **Human-centered computing** → *Haptic devices*.

Keywords

musical haptics, wearable, audio to haptic

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1 Introduction

Haptic feedback has increasingly been explored as a way to enrich music listening by allowing users to feel rhythmic and tonal

elements such as beats, dynamics, and melodies. Many existing systems generate vibrations directly from the audio frequency spectrum, which works well to convey low-level features such as tempo or intensity. However, such methods often fail to distinguish overlapping instruments in polyphonic music, suffering from the cocktail party problem in the vibrotactile domain [8]. This limitation reduces the clarity of musical structure, particularly in complex passages involving multiple instruments.

Recent advances in deep learning have made it possible to decompose a single audio track into distinct musical components, such as timbre and pitch. Building on this capability, we propose a novel approach to haptic music feedback. Our system separates individual instruments from a mixed track and maps each instrument's pitch to spatially distributed actuators. As a result, the method enhances clarity, intuitiveness, and comprehension of listener, providing an enhanced perception of individual instruments and understanding of musical structure.

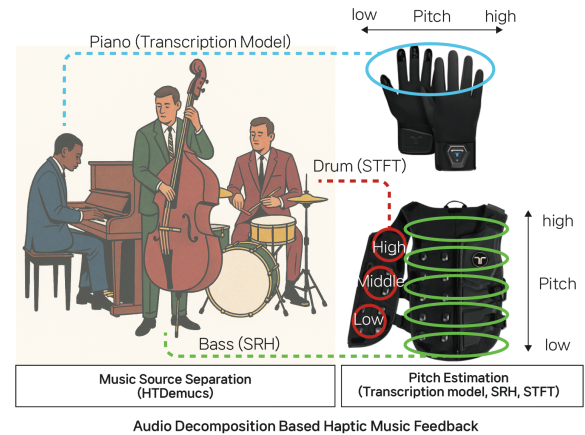


Figure 1: Configuration of the proposed audio decomposition-based haptic music feedback system. The system separates the sounds of piano, bass, and drums, and estimates their pitches individually. Based on the instrument type and pitch, it delivers vibrations to distinct spatial locations.

*Both authors contributed equally to this research.

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2 Implementation

2.1 Music Source Separation & Pitch Estimation

We selected the HTDemucs v4 model [9] for music source separation due to its strong performance in polyphonic audio, achieved through a hybrid architecture that leverages both time and frequency domain processing, integrated via a cross-domain Transformer encoder. This design enables the model to effectively isolate individual instruments even in complex musical mixtures.

To maximize the separation performance of HTDemucs, we constructed our system based on jazz trio music consisting of bass, piano, and drums. Each instrument stem was extracted using HTDemucs and then individually processed for pitch estimation using the following methods:

- **Piano:** We used a high-resolution piano transcription model [6] for pitch estimation. As piano music is polyphonic—often combining multiple notes for melodies and harmonies—robust polyphonic pitch estimation was essential.
- **Bass:** For bass, we adopted the Summation of Residual Harmonics (SRH) algorithm [3], optimized for the 10–180 Hz range. Bass lines are typically slow, monophonic, and low-frequency, making SRH suitable for accurate pitch detection in this range.
- **Drums:** As drums are percussive and lack definite pitch, we used Short-Time Fourier Transform (STFT) features to divide the drum track into low (kick), mid (snare and toms), and high (hi-hat and cymbals) categories based on frequency. These were separable through simple thresholding, which makes a lightweight STFT-based approach sufficient.

2.2 Haptic Feedback Design

Our haptic system comprises a vest (40 Eccentric Rotating Mass actuators, ERMs) and gloves (10 Linear Resonant Actuators, LRAs). To design an intuitive and meaningful haptic experience, each instrument was mapped to a body region based on its musical role, timbral characteristics, and its natural association with the human body. Each instrument track is mapped as follows:

- **Piano:** Leveraging users’ familiarity with piano playing, pitches are mapped to the five fingertips of each hand. A total of 24 semitone-spaced notes over 2 to 3 octaves are grouped into ten intervals, arranged left to right when the palms face downward—so that higher pitches activate the rightmost fingers. Notes outside this range are assigned to the nearest edge interval.
- **Bass:** Mapped to the back of the vest to reflect the role of the bass as a background / rhythmic foundation. Using five vertical positions, we discretize the pitch range (10–180 Hz) in 35–40 Hz intervals, activating all four actuators in a row simultaneously for a broad spread sensation.
- **Drums:** Mapped to the front of the vest (chest and abdomen). The high drum represents the top row, the middle drum the middle, and the low drum the bottom row. Each category is assigned a progressively wider horizontal row to convey its timbral weight (e.g., more actuators for the low drum).

3 Comparative Study

3.1 Procedure

To investigate how different approaches to haptic rendering affect the perception of music, we conducted a within-subject comparative study between two methods: one that uses Fast Fourier Transform to map frequency bands from a mixed audio signal to spatial locations (FFT-based) [1], and another that semantically separates instruments and maps their pitch and identity to distinct haptic positions (Audio Decomposition-based, AD-based).

To identify which feedback method better complemented different musical sections, we asked participants to listen to a jazz trio performance and press a “Like” button whenever the haptic feedback felt particularly impressive or well-aligned with the music. Each participant experienced both feedback methods in separate sessions, each consisting of a full playback of the music video Supernova performed by the SM Jazz Trio [9]. The button presses served as implicit, time-stamped indicators of perceived synchrony between the music and haptic sensations.

Immediately after each session, participants completed a short questionnaire evaluating their experience with the corresponding feedback method. Ratings were collected across five perceptual dimensions: clarity, intuitiveness, comprehension, richness, and immersion. These dimensions were selected based on prior research on musical aesthetics, immersion, and usability [2, 4, 5, 7]. Each dimension included three to four items rated on a 7-point Likert scale. Additionally, open-ended questions gathered participants’ subjective reflections on their musical experience.

A total of twelve participants (5 male, 7 female; aged 22–29) were recruited for this comparative study. Each participant experienced both conditions in counterbalanced order to control for order effects. The study was conducted in a controlled, single-blind setting to minimize experimental bias. The entire test lasted approximately 30 minutes, with each session consisting of approximately 5 minutes of music listening and 10 minutes of post-experience questionnaires.

3.2 Result

To examine which musical segments were preferred under each condition, we analyzed the temporal distribution of “Like” button presses. For each session, the number of button presses was normalized so that the total for each participant per method summed

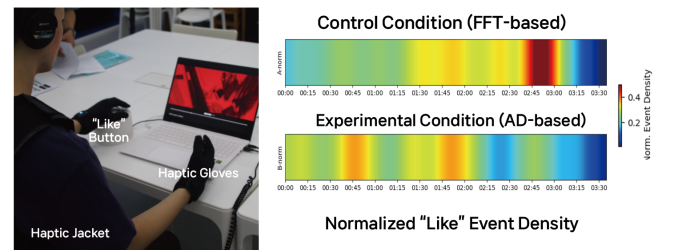


Figure 2: Experimental setup and results from the “Like” button log. Participants listened to jazz trio music and pressed a “Like” button whenever they perceived the haptic feedback to be well synchronized with the audio.

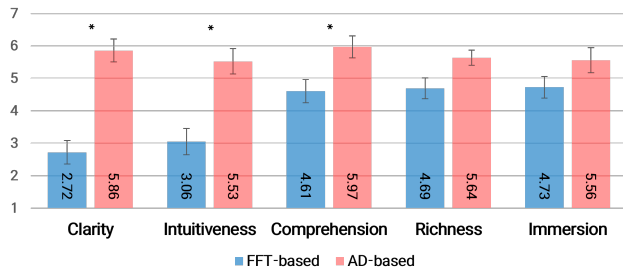


Figure 3: Survey results comparing the AD-based and FFT-based feedback methods across perceptual dimensions. Responses were rated on a 7-point Likert scale (1 for strongly disagree, 7 for strongly agree). Asterisk (*) denotes statistical significance at the $p < .05$ level.

to 1.0. This allowed us to compare the density of positive responses across time regardless of individual differences in response frequency. As shown in Figure 2, this approach enabled us to visualize participants’ preferred moments for each method using a heatmap, revealing differences in how each feedback strategy aligned with musical structure.

We focused on identifying which musical segments within each method attracted the most concentrated positive feedback. Within the FFT-based method, users tended to press the “Like” button more frequently during dense, full-instrument climax sections, suggesting a preference for its rendering in high-energy and ensemble-rich moments. In contrast, the AD-based method showed a higher density of positive responses during sequential layering passages, where individual instruments were introduced one at a time. As one participant noted, “*I could feel the buildup from bass to piano to drums through the haptic feedback—it made it more immersive.*” (P5) These patterns suggest that each method has unique strengths depending on the structural features of the music.

Figure 3 shows the results of the post-experience questionnaire. A paired t-test revealed that the AD-based method significantly outperformed the FFT-based method in terms of clarity, intuitiveness and comprehension ($p < .05$), indicating that it facilitated more intuitive and accurate recognition of individual instruments and supported better musical understanding. Although richness and immersion ratings were also higher for the AD-based method, these differences were not statistically significant. The feedback of the participants suggested that these two aspects were more influenced by individual listening styles: some preferred a holistic experience, as in “*I tend to focus on the overall mood rather than distinguishing between instruments.*” (P2), while others engaged with the music in a more analytical or embodied way, as in “*It felt as if I was playing the instrument with my own hands.*” (P9).

4 Conclusion and Future work

In this paper, we present an audio decomposition based approach to musical haptic feedback that leverages source separation and pitch estimation to deliver instrument-specific tactile cues. User studies showed that this method improved perceptual clarity and musical understanding. In terms of immersion, our method also

showed particular strength in sections where instruments entered sequentially, allowing users to experience the build-up of musical layers more vividly. In contrast, the conventional FFT-based method was effective in dense, high-energy passages, suggesting that each method offers distinct perceptual strengths depending on the musical structure. Building on these findings, our future work aims to develop a hybrid system that dynamically integrates AD- and FFT-based methods according to the real-time musical context, enabling more adaptive and immersive haptic music experiences.

Acknowledgments

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